

# TRIP13 localizes to synapsed chromosomes and functions as a dosage-sensitive regulator of meiosis

## Reviewed Preprint

Revised by authors after peer review.

## About eLife's process

## Reviewed preprint version 2

March 7, 2024 (this version)

## Reviewed preprint version 1

November 22, 2023

## Posted to preprint server

September 26, 2023

## Sent for peer review

September 8, 2023

Jessica Y. Chotiner, N. Adrian Leu, Fang Yang, Isabella G. Cossu, Yongjuan Guan, Huijuan Lin, P. Jeremy Wang 

Department of Biomedical Sciences, University of Pennsylvania School of Veterinary Medicine, Philadelphia, PA 19104, USA • College of Life Sciences, Capital Normal University, Beijing, China

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## Abstract

Meiotic progression requires coordinated assembly and disassembly of protein complexes involved in chromosome synapsis and meiotic recombination. The AAA+ ATPase TRIP13 and its orthologue Pch2 are instrumental in remodeling HORMA domain proteins. Meiosis-specific HORMAD proteins are associated with unsynapsed chromosome axes but depleted from the synaptonemal complex (SC) of synapsed chromosome homologues. Here we report that TRIP13 localizes to the synapsed SC in early pachytene spermatocytes and to telomeres throughout meiotic prophase I. Loss of TRIP13 leads to meiotic arrest and thus sterility in both sexes. *Trip13*-null meiocytes exhibit abnormal persistence of HORMAD1 and HOMRAD2 on synapsed SC and chromosome asynapsis that preferentially affects XY and centromeric ends. These findings confirm the previously reported phenotypes of the *Trip13* hypomorph alleles. *Trip13* heterozygous (*Trip13*<sup>+/−</sup>) mice also exhibit meiotic defects that are less severe than the *Trip13*-null mice, showing that TRIP13 is a dosage-sensitive regulator of meiosis. Localization of TRIP13 to the synapsed SC is independent of SC axial element proteins such as REC8 and SYCP2/SYCP3. The N- or C-terminal FLAG-tagged TRIP13 proteins are functional and recapitulate the localization of native TRIP13 to SC and telomeres in knockin mice. Therefore, the evolutionarily conserved localization of TRIP13/Pch2 to the synapsed chromosomes provides an explanation for dissociation of HORMA domain proteins upon chromosome synapsis in diverse organisms.

## eLife assessment

This **important** study defined the physiological function of a conserved meiosis factor during murine spermatogenesis. The genetic and cellular biological evidence supporting the conclusion is **convincing**. This work will be of broad interest to cell biologists, geneticists, and reproductive biologists.

## Introduction

Meiosis is a specialized cell division program that generates haploid gametes from diploid germ cells. During meiotic prophase I, homologous chromosomes undergo pairing, synapsis, and recombination (Handel and Schimenti, 2010; Zickler and Kleckner, 2015). Chromosome synapsis requires the formation of the synaptonemal complex (SC). The SC is a proteinaceous structure comprised of two lateral/axial elements, transverse filaments, and a central element (Page and Hawley, 2004). Meiotic recombination begins with generation of DNA double strand breaks (DSBs) and ends with crossover formation through repair of DSBs (Hunter, 2015). Chromosome synapsis and meiotic recombination are interdependent in many species including yeast and mouse. Meiotic recombination not only increases genetic diversity in gametes at the organism level but also is essential for faithful chromosome segregation during the first meiotic cell division. Therefore, defects in meiosis are leading causes of aneuploidy, birth defect, infertility, and pregnancy loss.

The progression and completion of synapsis and recombination are monitored by a surveillance mechanism called meiotic checkpoint (Roeder and Bailis, 2000). Meiotic checkpoint proteins were extensively studied in yeast. Many meiotic processes are highly conserved, and studies of meiosis in yeast have been foundational for our understanding of mammalian meiosis. In yeast, Pch2, an AAA+ ATPase, is a checkpoint protein, because it is necessary for pachytene arrest in the absence of the SC protein Zip1 or the meiosis-specific DSB repair protein Dmc1 (Herruzo et al., 2021; San-Segundo and Roeder, 1999). Pch2 is a hexameric ring ATPase and remodels the HORMA (Hop1, Rev7, and MAD2) domain protein Hop1 in a nucleotide-dependent manner (Chen et al., 2014). While Pch2 is mainly located in the nucleolus, Pch2 also localizes to the synapsed SC as foci at the pachytene stage and is essential for removing Hop1 from chromosome axes (Chen et al., 2014; Joshi et al., 2009; San-Segundo and Roeder, 1999). Mechanistically, Pch2 promotes phosphorylation of Hop1, which activates the Mek1 kinase and the subsequent checkpoint cascade (Herruzo et al., 2016; Raina and Vader, 2020).

TRIP13, the mammalian orthologue of yeast Pch2, is essential for completion of meiotic recombination in mouse (Li and Schimenti, 2007; Roig et al., 2010). Mutations in human *TRIP13* predispose to Wilms tumor in children or cause infertility in women (Yost et al., 2017; Zhang et al., 2020). HORMAD1 and HORMAD2, mammalian meiosis-specific HORMA domain proteins, localize to chromosome axis at leptotene and zygotene stages but are depleted from synapsed chromosomes except the largely unsynapsed XY chromosomes at the pachytene stage (Fukuda et al., 2010; Xu et al., 2019). TRIP13 facilitates removal of HORMAD proteins from synapsed chromosome axis (Roig et al., 2010; Wojtasz et al., 2009). SKP1, an essential component of the SCF (Skp1-Cullin 1-F-box protein) ubiquitin E3 ligase, is required not only for depleting HORMAD1 and HORMAD2 from synapsed chromosome axes but also for restricting the accumulation of HORMAD proteins on unsynapsed axes (Guan et al., 2020; Guan et al., 2022). Therefore, reorganization of HORMA domain proteins on meiotic chromosome axes is regulated by both TRIP13 and SKP1. As a conserved mechanism, Pch2/TRIP13 remodels HORMA domain proteins through engagement of their N-terminal regions (Prince and Martinez-Perez, 2022). MAD2, a spindle assembly checkpoint HORMA domain protein, exists in two fold states: closed and open (Luo et al., 2004; Sironi et al., 2002). In the closed state, the so-called safety belt of the HORMA domain wraps around the HORMAD protein-binding motif “closure motif” in the partner protein and thus traps the partner. In the open state, TRIP13 engagement changes conformation of the HORMA domain, resulting in foldback of the safety belt on the closure motif-binding region and thus release of its partner protein. The safety belt mechanism turns out to be a shared feature of HORMA domain proteins (Kim et al., 2014; West et al., 2019; West et al., 2018; Ye et al., 2017).

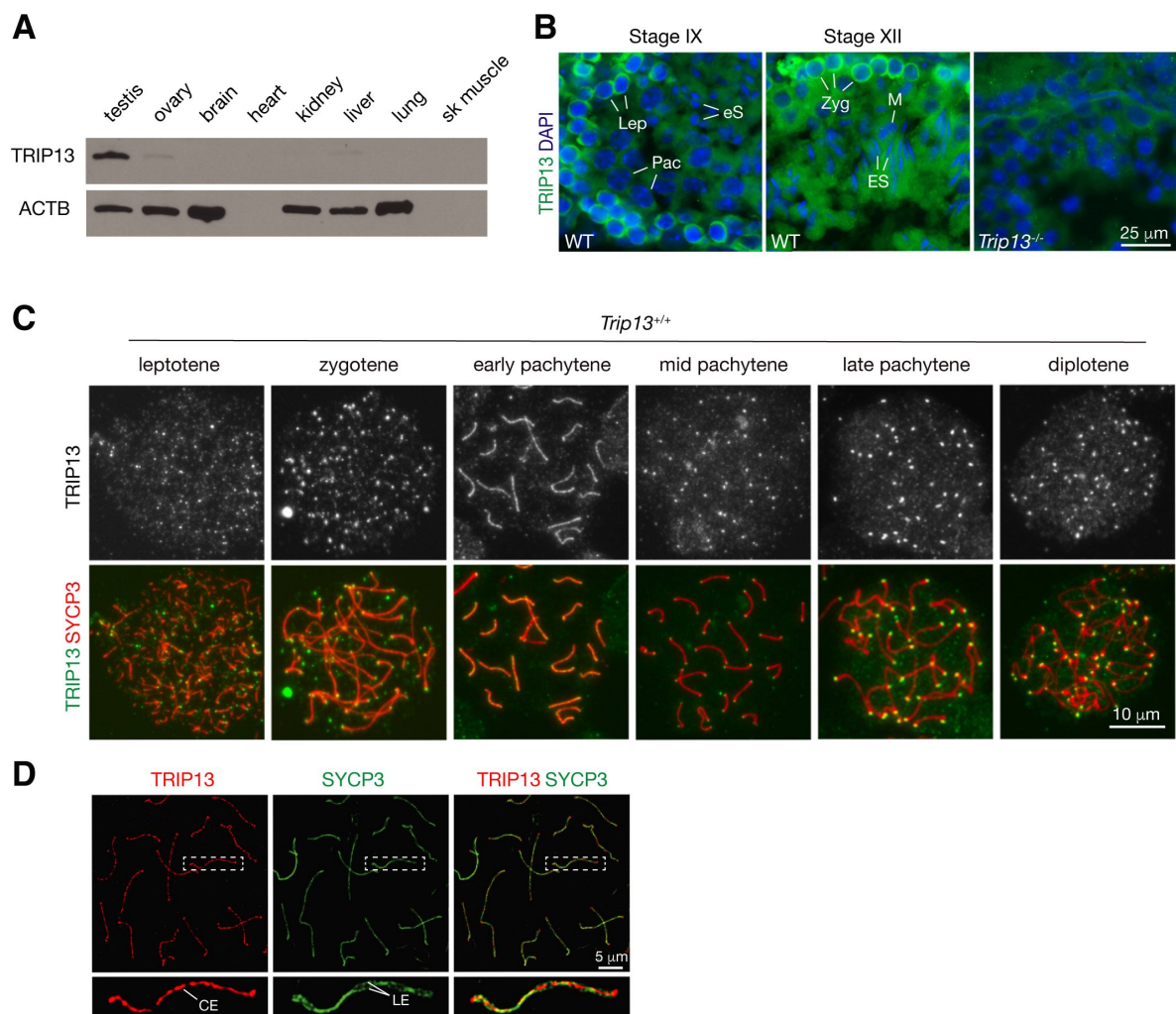
Unlike yeast Pch2, TRIP13 does not appear to function in the meiotic checkpoint in mouse (Pacheco et al., 2015 [↗](#)). Interestingly, HORMAD2 functions as a meiotic checkpoint protein for surveillance of meiotic defects in female meiosis (Kogo et al., 2012a [↗](#); Rinaldi et al., 2017 [↗](#); Wojtasz et al., 2012 [↗](#)). HORMAD2-deficient males are sterile but females are fertile. HORMAD2 deficiency rescues the oocyte loss in *Spo11*-null or *Trip13* mutant females but not in *Dmc1*-null females, suggesting that HORMAD2 is a component of the meiotic checkpoint in females. HORMAD1 is required for fertility in both sexes (Kogo et al., 2012b [↗](#); Shin et al., 2010 [↗](#); Stanzione et al., 2016 [↗](#)). Both HORMAD1 and HORMAD2 localize prominently to the XY chromosome axes in pachytene spermatocytes and regulate meiotic sex chromatin inactivation (MSCI) (Kogo et al., 2012a [↗](#); Shin et al., 2010 [↗](#); Wojtasz et al., 2012 [↗](#)).

Studies of mice with hypomorphic *Trip13* mutations show that TRIP13 is required for meiotic progression (Li and Schimenti, 2007 [↗](#); Roig et al., 2010 [↗](#)). Two different hypomorphic *Trip13* mouse lines have been studied: one “moderate” allele that exhibited defective meiotic recombination and one “severe” allele that displayed defects in meiotic recombination and chromosome synapsis. The variance in phenotype between the hypomorphic mouse lines indicates that even partial depletion of *Trip13* can interrupt meiosis. We sought to further investigate the meiotic function of TRIP13. By immunofluorescence, we found that TRIP13 localized to the synaptonemal complex in early pachytene spermatocytes and to telomeres throughout meiotic prophase I. The TRIP13 localization pattern on meiotic chromosomes was further confirmed by analysis of FLAG-tagged TRIP13 in two knockin mouse lines. To determine the effect of complete loss of TRIP13, we characterized *Trip13*-null mice. While the *Trip13*-null phenotype was largely comparable to the phenotype of the severe *Trip13* hypomorphic allele, we found that TRIP13 is a dosage-sensitive regulator of meiosis. Finally, our results show that the localization of TRIP13 to the SC is independent of axial element proteins.

## Results

### TRIP13 localizes to meiotic chromosomes in prophase I spermatocytes

We assessed the expression of TRIP13 in a panel of adult mouse tissues. TRIP13 was primarily expressed in testis but detected at low levels in ovary and liver (**Figure 1A** [↗](#)). In the testis, TRIP13 was prominent in the cytoplasm of primary spermatocytes, especially leptotene and zygotene cells (**Figure 1B** [↗](#)). Given the expression of TRIP13 in spermatocytes, we examined its localization on meiotic chromosomes by immunostaining of surface spread nuclei of prophase I spermatocytes (**Figure 1C** [↗](#)). Previously, TRIP13 was reported to localize to telomeres in spermatocytes (Gomez et al., 2019 [↗](#)). Indeed, we found that TRIP13 localized to telomeres from leptotene to diplotene cells (**Figure 1C** [↗](#)). Strikingly, we found that, in addition to telomeres, TRIP13 localized to the synaptonemal complex (SC) in early pachytene spermatocytes but disappeared from the SC in the mid to late pachytene spermatocytes (**Figure 1C** [↗](#)). The synaptonemal complex consists of two lateral elements and one central element. The central region is comprised of the transverse filament protein SYCP1 and central element proteins, which appear upon synapsis. Given the novel localization of TRIP13 to the SC, we performed confocal immunofluorescent microscopy with super-resolution deconvolution to determine the location of TRIP13 within the SC. In contrast with the lateral element marker SYCP3, TRIP13 localized to the SC central region (**Figure 1D** [↗](#)). We examined the localization of TRIP13 in oocytes. TRIP13 localized strongly to meiotic telomeres in all stages of prophase I, but not as filaments on the SC (**Figure 1 - figure supplement 1** [↗](#)). These findings are consistent with the genetic requirement of TRIP13 in meiosis in both sexes (Li and Schimenti, 2007 [↗](#); Roig et al., 2010 [↗](#)).



**Figure 1.**

TRIP13 localizes to the synaptonemal complex and telomeres in spermatocytes. (A) Western blot analysis of TRIP13 in adult mouse tissues. Heart and skeletal muscle lack ACTB. (B) Immunofluorescence of TRIP13 in sections of 3-month-old wild type and *Trip13<sup>-/-</sup>* testes. Lep, leptotene; Zyg, zygotene; Pac, pachytene; eS, elongating spermatids; ES, elongated spermatids. (C) Immunofluorescence of TRIP13 in spread nuclei of spermatocytes from wild type P20 testes. (D) Super-resolution localization of TRIP13 to the central element (CE) but not lateral element (LE) of the synaptonemal complex at early pachytene stage. The enlarged view of the boxed chromosome is shown at the bottom.

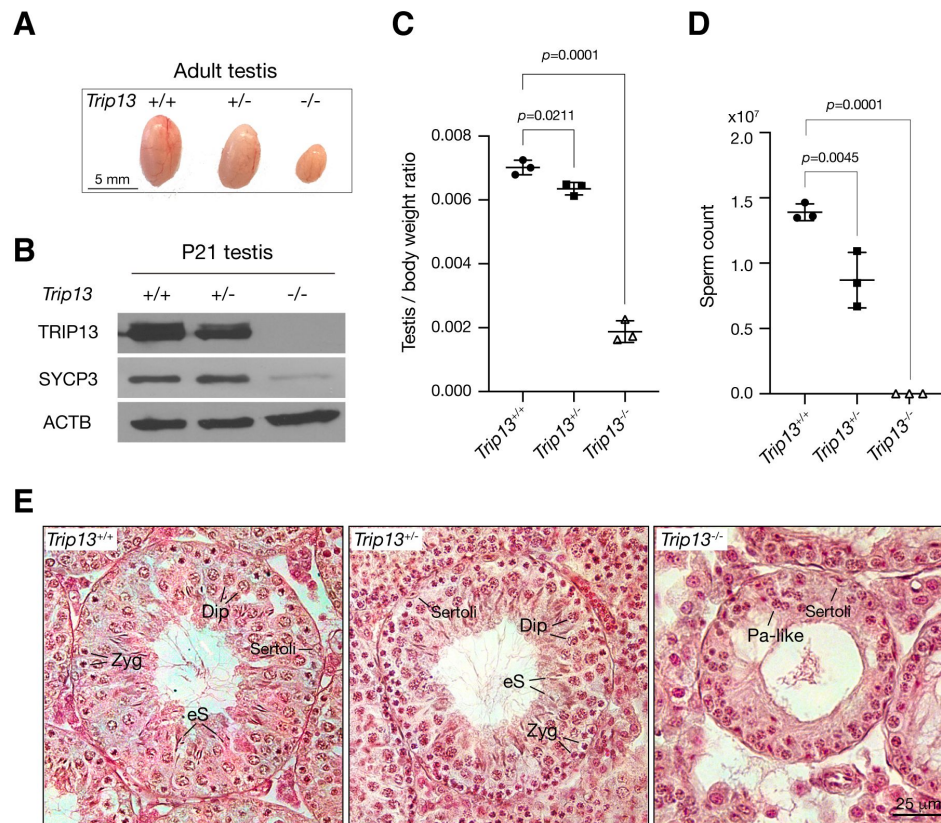
## Global loss of *Trip13* causes meiotic arrest

Two previous studies utilizing two *Trip13* hypomorphic (gene trap) alleles, one moderate and one severe, revealed a role for TRIP13 in meiotic recombination (Li and Schimenti, 2007 [\[1\]](#); Roig et al., 2010 [\[2\]](#)). However, while the *Trip13* moderate allele showed mostly intact chromosomal synapsis, the severe allele displayed chromosomal unsynapsis preferentially at chromosome ends (Li and Schimenti, 2007 [\[1\]](#); Roig et al., 2010 [\[2\]](#)). Upon close examination, we also found unsynapsed ends in spermatocytes with the moderate *Trip13* mutant (Guan et al., 2020 [\[3\]](#)). To rigorously ascertain the role of TRIP13 in meiosis, we generated *Trip13*-null mutants using frozen sperm from *Trip13*<sup>+/-</sup> males with a knockout allele from the International Mouse Phenotyping Consortium (IMPC). The mouse *Trip13* gene consists of 13 exons. This new *Trip13* mutant allele harbors a deletion of a 24-kb region including all 13 exons and thus is expected to be null. We verified this deletion by PCR and sequencing. Interbreeding of *Trip13*<sup>+/-</sup> mice produced fewer *Trip13*<sup>-/-</sup> offspring than expected: *Trip13*<sup>+/+</sup>, 80; *Trip13*<sup>+/-</sup>, 220; *Trip13*<sup>-/-</sup>, 51 ( $\chi^2=27$ ,  $p=0.0001$ ). The body weight of 2-3 month-old males was not significantly different between wild type (24.3±2.8 g, n=5) and *Trip13*<sup>-/-</sup> mice (22.8±1.7 g, n=5,  $p=0.3$ , Student's *t*-Test). The *Trip13*<sup>-/-</sup> testis was much smaller than *Trip13*<sup>+/-</sup> testis (**Figure 2A** [\[4\]](#)). Western blot analysis showed that TRIP13 was present in reduced abundance in *Trip13*<sup>+/-</sup> testis and not detected in *Trip13*<sup>-/-</sup> testis, demonstrating that this mutant allele is null (**Figure 2B** [\[4\]](#)). The abundance of SYCP3, a component of the synaptonemal complex, was also reduced in *Trip13*<sup>-/-</sup> testis, suggesting a partial depletion of meiotic cells (**Figure 2B** [\[4\]](#)). The testis weight of adult *Trip13*<sup>-/-</sup> males was reduced by 74% in comparison with the wild type males (**Figure 2C** [\[4\]](#)). Adult *Trip13*<sup>-/-</sup> males lacked sperm in the epididymis (**Figure 2D** [\[4\]](#)). Histological analysis showed that while spermatocytes in all stages of meiosis were present in adult *Trip13*<sup>+/+</sup> and *Trip13*<sup>+/-</sup> testes, *Trip13*<sup>-/-</sup> testis displayed complete meiotic arrest, evidenced by the presence of early spermatocytes and a lack of secondary spermatocytes, round spermatids, and mature sperm (**Figure 2E** [\[4\]](#)). *Trip13*<sup>-/-</sup> females were also sterile. The adult *Trip13*<sup>-/-</sup> ovary was very small and showed a complete loss of oocytes (**Figure 2 - figure supplement 1** [\[4\]](#)). These observations are similar to the meiotic arrest phenotype observed in the hypomorphic *Trip13* mouse mutants (Li and Schimenti, 2007 [\[1\]](#); Roig et al., 2010 [\[2\]](#)).

## Defects in chromosomal synapsis in *Trip13*-deficient spermatocytes

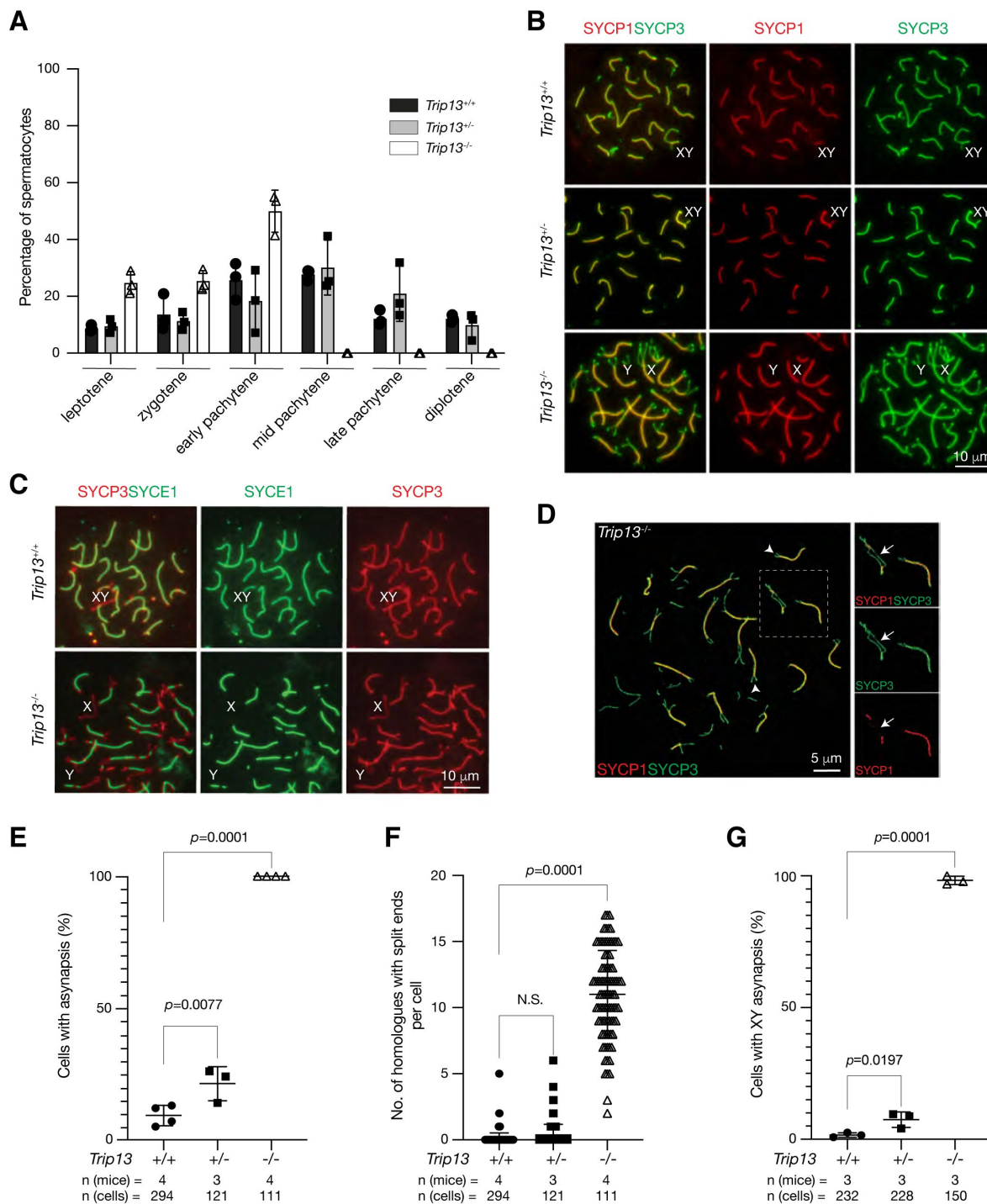
We investigated the effect of *Trip13* deficiency on meiotic progression. The transverse filament/central element protein SYCP1 and the lateral element protein SYCP3 were used to determine the stage of prophase I spermatocytes. While the P20 wild type testis contained spermatocytes from leptotene through diplotene stages, about half of the *Trip13*<sup>-/-</sup> spermatocytes were in the early pachytene stage and no cells were found at later stages of prophase I, showing meiotic blockade at the early pachytene stage in *Trip13*<sup>-/-</sup> testes (**Figure 3A** [\[4\]](#)). The early pachytene-like spermatocytes from *Trip13*<sup>-/-</sup> testes contained unsynapsed chromosomal ends (**Figure 3B** [\[4\]](#)). Mouse centromeres are telocentric. Co-staining with the centromere marker CREST showed that 94% of unsynapsed ends were centromeric ends (**Figure 4A** [\[4\]](#)). While XY chromosomes were synapsed at the pseudoautosomal regions and thus were connected in wild type and *Trip13*<sup>+/-</sup> pachytene spermatocytes, they were separate in *Trip13*<sup>-/-</sup> pachytene-like spermatocytes (**Figure 3B** [\[4\]](#) and **Figure 3C** [\[4\]](#)). We found that SYCE1, a component of the SC central element, localized to the synapsed SC but not to unsynapsed regions in *Trip13*<sup>-/-</sup> pachytene-like cells (**Figure 3C** [\[4\]](#)). Confocal microscopy with super-resolution deconvolution confirmed that many homologous chromosomes in *Trip13*<sup>-/-</sup> spermatocytes had split ends and some had regions of interstitial asynapsis (**Figure 3D** [\[4\]](#)). We quantified these meiotic defects in spermatocytes: chromosomal asynapsis (**Figure 3E** [\[4\]](#)), the number of asynapsed ends (**Figure 3F** [\[4\]](#)), and the extent of XY asynapsis (**Figure 3G** [\[4\]](#)). Previous studies also reported synaptic defects in spermatocytes from *Trip13* hypomorph mutants (Li and Schimenti, 2007 [\[1\]](#); Roig et al., 2010 [\[2\]](#)). Here we found that global loss of *Trip13* caused similar defects in autosomal synapsis but a more severe defect in sex chromosome synapsis.





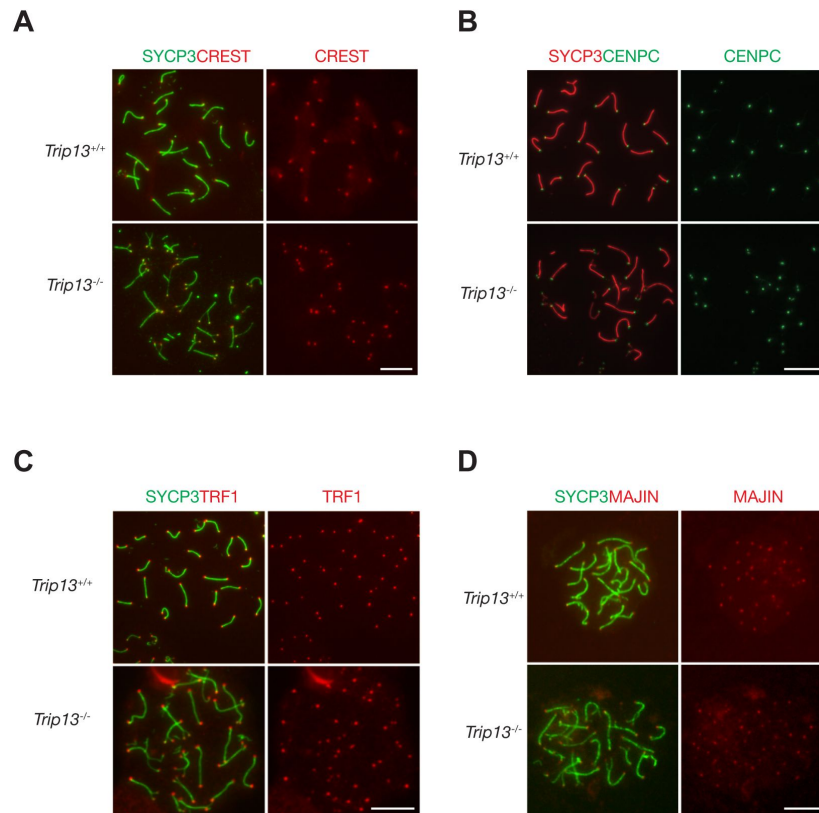
**Figure 2.**

Loss of *Trip13* leads to meiotic arrest in males. (A) Image of testes from 2 to 3 month-old mice. (B) Western blot analysis of TRIP13 in P21 testes. SYCP3 serves as a meiosis-specific marker. ACTB serves as a loading control. (C) Testis to body weight ratio of 2 to 3 month-old mice.  $n = 3$  males. Statistics, one-Way ANOVA. (D) Sperm count of 2 to 3 month-old *Trip13* $^{+/+}$  and *Trip13* $^{-/-}$  males.  $n = 3$  males. Statistics, one-Way ANOVA. (E) Histological analysis of 2-month-old testes. Sertoli, Sertoli cell; Zyg, zygotene; Pa-like, pachytene-like; Dip, diplotene; eS, elongating spermatids.



**Figure 3.**

*Trip13* is required for chromosomal synapsis in males. (A) Composition of prophase I spermatocytes in P20 testes. Three males per genotype were analyzed by nuclear spread analysis. Total number of spermatocytes counted: *Trip13*<sup>+/+</sup>, 1038 cells; *Trip13*<sup>+/-</sup>, 803 cells; *Trip13*<sup>-/-</sup>, 406 cells. (B) Immunofluorescence of SYCP1 and SYCP3 in spread nuclei of pachytene spermatocytes from P20 testes. (C) Immunofluorescence of SYCE1 and SYCP3 in spread nuclei of pachytene spermatocytes from P20 testes. (D) Super-resolution confocal microscopy of a *Trip13*<sup>-/-</sup> spermatocyte from P20 testis. Immunostaining was performed for SYCP1 and SYCP3. Arrowheads indicate end asynapsis. Arrow indicates interstitial asynapsis. (E) Percentage of early pachytene cells from P19-20 testes with asynapsed chromosomes across three genotypes. (F) Number of homologous chromosomes with end asynapsis per cell in P19-20 testes. (G) Percentage of early pachytene cells from P20 testes with asynapsed XY chromosomes per mouse. The *p* values are indicated in graphs. Statistics (E-G), one-way ANOVA.



**Figure 4.**

Normal localization of centromere and telomere markers in *Trip13*-deficient spermatocytes from juvenile mice. (A-D) Immunofluorescent analysis of centromere and telomere markers in *Trip13*<sup>+/+</sup> and *Trip13*<sup>-/-</sup> pachytene spermatocytes: CREST (A), CENPC (B), TRF1 (C), and MAJIN (D). SYCP3 labels the lateral elements of the synaptonemal complex. Scale bars, 10 μm.



## TRIP13 is a dosage-sensitive regulator of meiosis

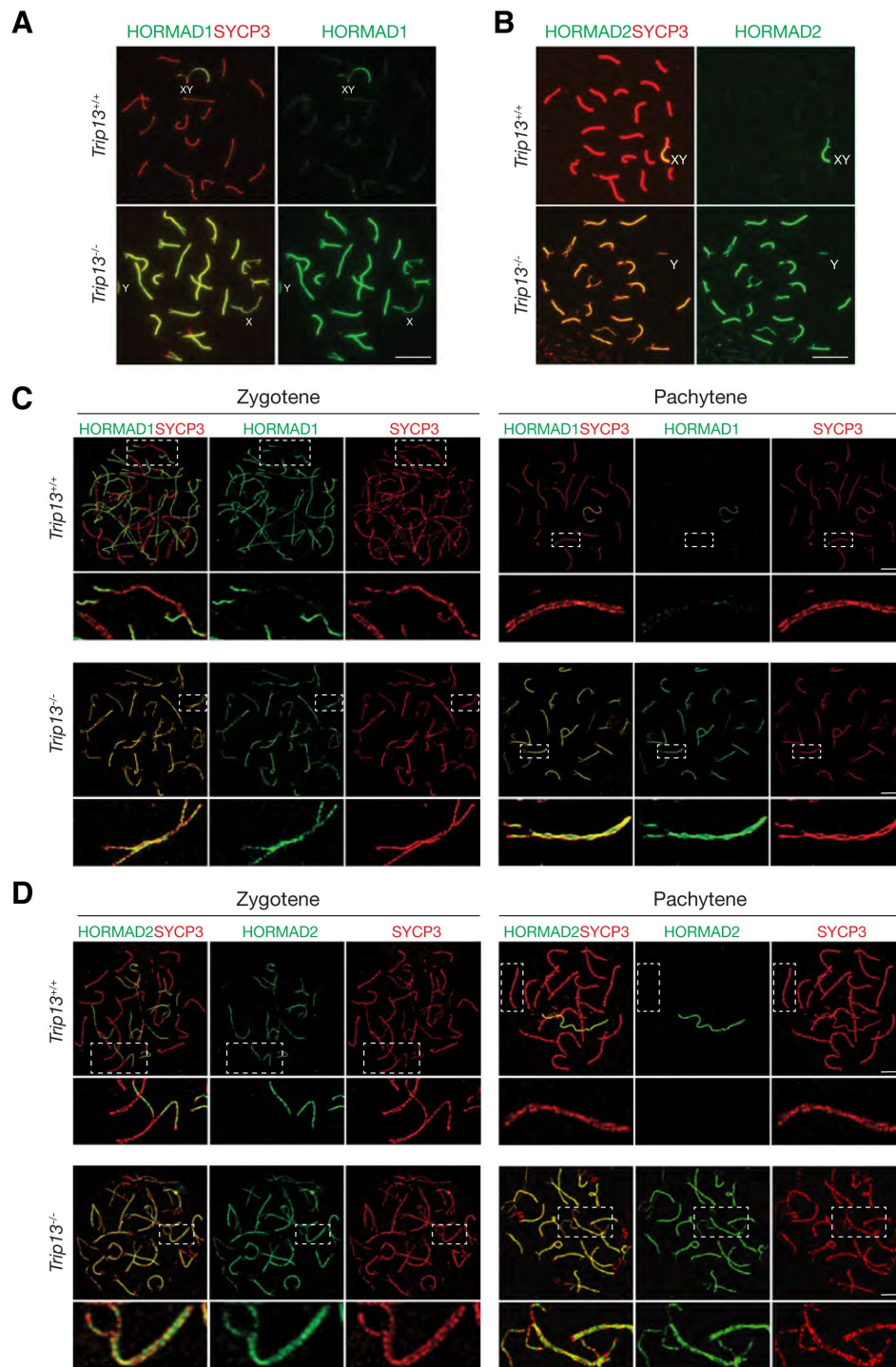
Although *Trip13*<sup>+/-</sup> males were fertile, their testis weight and sperm count were significantly reduced in comparison with wild type (**Figure 2C-D**). The percentage of spermatocytes with defects in synapsis was higher in *Trip13*<sup>+/-</sup> males than wild type (**Figure 3E**). In addition, the percentage of spermatocytes with asynapsed XY chromosomes was significantly higher in *Trip13*<sup>+/-</sup> males than wild type (**Figure 3G**). As defects in synapsis can activate the meiotic checkpoint and thus apoptosis of affected spermatocytes, the increased synaptic defects likely caused the decrease in sperm output in *Trip13*<sup>+/-</sup> males. These results demonstrate that TRIP13 regulates meiosis in a dosage-dependent manner.

## Telomere and centromere proteins localize normally in *Trip13*-deficient spermatocytes

TRIP13 localizes to telomeres and loss of TRIP13 causes peri-centromeric/telomeric asynapsis. Therefore, we asked whether telomere or centromere proteins were affected in *Trip13*<sup>-/-</sup> spermatocytes. Chromosome spreads were immunostained for two centromere markers, CREST and CENPC (**Figure 4A-B**). CENPC is essential for recruiting kinetochore proteins to the centromere (Klare et al., 2015; Kwon et al., 2007). CREST and CENPC still localized to centromeres in *Trip13*<sup>-/-</sup> spermatocytes. Because many *Trip13*<sup>-/-</sup> spermatocytes had split ends, two CREST or two CENPC foci were observed at the split ends. To assess the telomere, spermatocytes were immunostained for TRF1 and MAJIN (**Figure 4C-D**). In meiotic cells, telomeres contain canonical telomere proteins such as TRF1/TRF2 and a meiosis-specific complex (MAJIN, TERB1, TERB2, and SUN1) (Shibuya et al., 2015). TRF1 is a telomere protein expressed in both somatic and germ cells (Karlseider et al., 2003; Long et al., 2017; Shibuya et al., 2014). MAJIN is a component of the meiosis-specific telomere complex that is important for telomere attachment to the inner nuclear membrane (Shibuya et al., 2015). Both TRF1 and MAJIN localized to telomeres in *Trip13*<sup>-/-</sup> spermatocytes (**Figure 4C-D**). As expected, two TRF1 or two MAJIN foci were observed at split ends of chromosomes in *Trip13*<sup>-/-</sup> spermatocytes (**Figure 4C-D**). ANKRD31 and REC114 interact with each other and both localize to the pseudoautosomal region (PAR) of XY chromosomes in early pachytene spermatocytes to ensure XY recombination (Boekhout et al., 2019; Papanikos et al., 2019). In addition to two foci on autosomes, REC114 localized as one focus on the PAR in wild type pachytene spermatocytes. REC114 still formed foci (one per chromosome) on the unsynapsed X and Y chromosomes in *Trip13*<sup>-/-</sup> pachytene-like spermatocytes (**Figure 4 - figure supplement 1**). Taken together, these results suggest that TRIP13 is not required for recruitment of these centromere or telomere proteins.

## TRIP13 is required to evict HORMAD1 and HORMAD2 from synapsed autosomes

In wild type meiotic cells, HORMAD1 and HORMAD2 localize to unsynapsed and desynapsed chromosomes (Fukuda et al., 2010; Kogo et al., 2012a; Shin et al., 2010; Wojtasz et al., 2012). Thus, HORMAD1 and HORMAD2 localized to the largely unsynapsed XY but not to synapsed autosomal SCs in wild type cells (**Figure 5A-B**). TRIP13 is essential for removing meiotic HORMADs from the chromosome axes, a function that is conserved in yeast, worms, and mammals (Wojtasz et al., 2009). We confirmed that HORMAD1 and HORMAD2 remained on the synapsed autosomes in *Trip13*<sup>-/-</sup> pachytene cells (**Figure 5A-B**). HORMAD1 and HORMAD2 localize to the interior of the lateral elements in the SC (Xu et al., 2019). Confocal microscopy with super-resolution deconvolution revealed that HORMAD1 and HORMAD2 localized to the lateral elements of the synapsed autosomes in *Trip13*<sup>-/-</sup> spermatocytes (**Figure 5C-D**). These results were consistent with accumulation of HORMAD1/2 in *Trip13* hypomorphic mutant spermatocytes (Roig et al., 2010; Wojtasz et al., 2009). Our analysis of *Trip13*-null mutant confirmed that TRIP13 is essential for HORMAD1/2 removal from the synapsed SCs.



**Figure 5.**

HORMAD1 and HORMAD2 accumulate on the lateral elements of synapsed autosomes in *Trip13<sup>-/-</sup>* spermatocytes from juvenile (P19-21) mice. (A, B) Immunofluorescence of HORMAD1 (A) and HORMAD2 (B) in pachytene spermatocytes. (C, D) Super-resolution imaging of HORMAD1 and HORMAD2 in zygotene and pachytene spermatocytes. Enlarged views of the boxed chromosomes are shown below. Scale bars: 10  $\mu$ m (A, B), 5  $\mu$ m (C, D).

## Localization of TRIP13 to SC is independent of axial element components

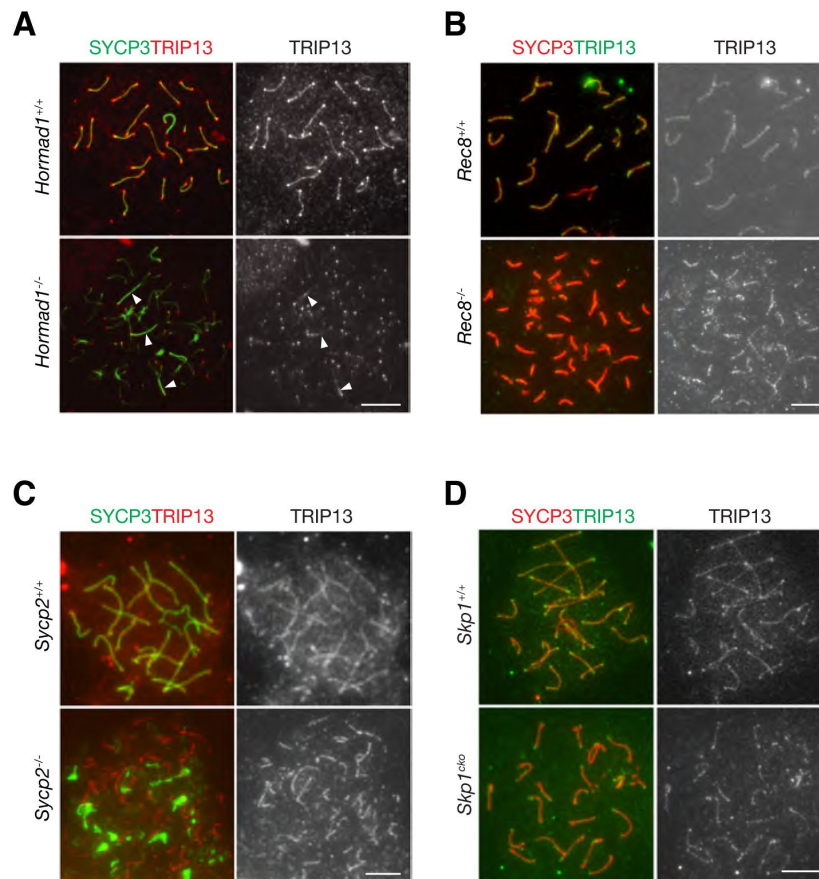
TRIP13 localizes to the SC in early pachytene spermatocytes. We sought to address what proteins might recruit TRIP13 to the SC. We examined the role of HORMAD1, REC8, SYCP2, and SKP1 in TRIP13 localization in spermatocytes using respective knockout mice (**Figure 6**). We first examined the localization of TRIP13 in *Hormad1*<sup>-/-</sup> spermatocytes. *Hormad1*<sup>-/-</sup> spermatocytes exhibit limited synapsis (Daniel et al., 2011; Kogo et al., 2012b; Shin et al., 2010). In both wild type and *Hormad1*<sup>-/-</sup> cells, TRIP13 localized to the synapsed regions and to the telomeres (**Figure 6A**). This result shows that HORMAD1 is not required for localization of TRIP13 to the SC.

REC8, a meiosis-specific cohesin, promotes synapsis between homologs and inhibits synapsis between sister chromatids (Xu et al., 2005). In the absence of REC8, synapsis occurs between sister chromatids rather than homologous chromosomes as previously shown by super-resolution imaging (Guan et al., 2020). In *Rec8*<sup>-/-</sup> spermatocytes, TRIP13 localized to synapsed sister chromatids (**Figure 6B**). To further probe TRIP13 recruitment to the SC, a mouse line expressing a truncated SYCP2 was used (Yang et al., 2006). SYCP2 and SYCP3 are essential components of the axial/lateral elements of the SC and interact with each other. The mutant used in this study (referred to as *Sycp2*<sup>-/-</sup>) expresses a truncated SYCP2 protein that lacks the C-terminal coiled-coil domain necessary for binding to SYCP3. Thus, SYCP3 failed to localize to the axial elements and formed aggregates in *Sycp2*<sup>-/-</sup> spermatocytes (**Figure 6C**). Nevertheless, *Sycp2*<sup>-/-</sup> spermatocytes formed short stretches of synapsis as previously demonstrated by electron microscopy and SYCP1 immunofluorescence (Yang et al., 2006). TRIP13 localized as filaments in *Sycp2*<sup>-/-</sup> spermatocytes, strongly suggesting that its localization is independent of SYCP3 and the C-terminus of SYCP2 (**Figure 6C**).

Previous work has shown that SKP1, a key component of the SKP1, Cullin, F-box (SCF) complex E3 ligase, is important for HORMAD removal and chromosomal synapsis (Guan et al., 2020; Guan et al., 2022). SKP1 localizes to synapsed regions in meiotic germ cells and specifically to the lateral elements in the SC (Guan et al., 2020). We found that TRIP13 still localized to the synapsed regions in *Skp1*<sup>CKO</sup> (conditional knockout) spermatocytes (**Figure 6D**). Taken together, these results show that the SC localization of TRIP13 is independent of HORMAD1, REC8, SYCP2, SYCP3, and SKP1, which all localize to the SC lateral elements. Such a finding is consistent with the localization of TRIP13 to the SC central region (**Figure 1D**).

## FLAG-tagged TRIP13 proteins are functional

In order to investigate the mechanism of TRIP13 recruitment to the SC, we generated 3×FLAG-TRIP13 (N-terminal tag) and TRIP13-3×FLAG (C-terminal tag) mice through the CRISPR/Cas9-mediated genome editing approach (**Figure 7 - figure supplement 1**). Two different types of tagged mice were generated in case that one fusion protein is not functional. Both alleles were transmitted through the germline of the founder mice. Western blot showed that TRIP13 existed in two isoforms in wild type (non-tagged) testis (**Figure 7A**). The major TRIP13 isoform was 50 kDa. The minor isoform was slightly larger than 50 kDa. Western blot analysis showed that the FLAG tagged TRIP13 fusion proteins were present in both FLAG-tagged testes. Both FLAG-tagged TRIP13 fusion proteins also existed in two isoforms in the homozygous testes, suggesting that they corresponded to the two wild type isoforms. The 3×FLAG-TRIP13 proteins were apparently slightly larger, possibly due to the linker in the N-terminally tagged proteins. The nature and physiological significance of these two isoforms were not clear, but could be due to alternative splicing or post-translational modification. Immunofluorescence analysis using anti-FLAG antibody showed that the FLAG-tagged TRIP13 proteins, like wild type TRIP13, localized to both telomeres and synaptonemal complex in pachytene spermatocytes from both FLAG/FLAG homozygous testes.



**Figure 6.**

Localization of TRIP13 to the SC is independent of individual axial element components. (A) Immunofluorescent analysis of TRIP13 in *Hormad1*<sup>-/-</sup> spermatocytes from 2-month-old mice. (B) Immunofluorescent analysis of TRIP13 in *Rec8*<sup>-/-</sup> spermatocytes from 2-month-old mice. (C) Immunofluorescent analysis of TRIP13 in *Sycp2*<sup>-/-</sup> spermatocytes from 2-month-old mice. (D) Immunofluorescent analysis of TRIP13 in *Skp1*<sup>cko</sup> spermatocytes from 2-month-old mice. Scale bars, 10 μm.



(**Figure 7B** [↗](#)). Importantly, both N- and C-terminally tagged homozygous mice were fertile. These results demonstrate that both N- and C-terminal tagged TRIP13 proteins localize normally and are functional.

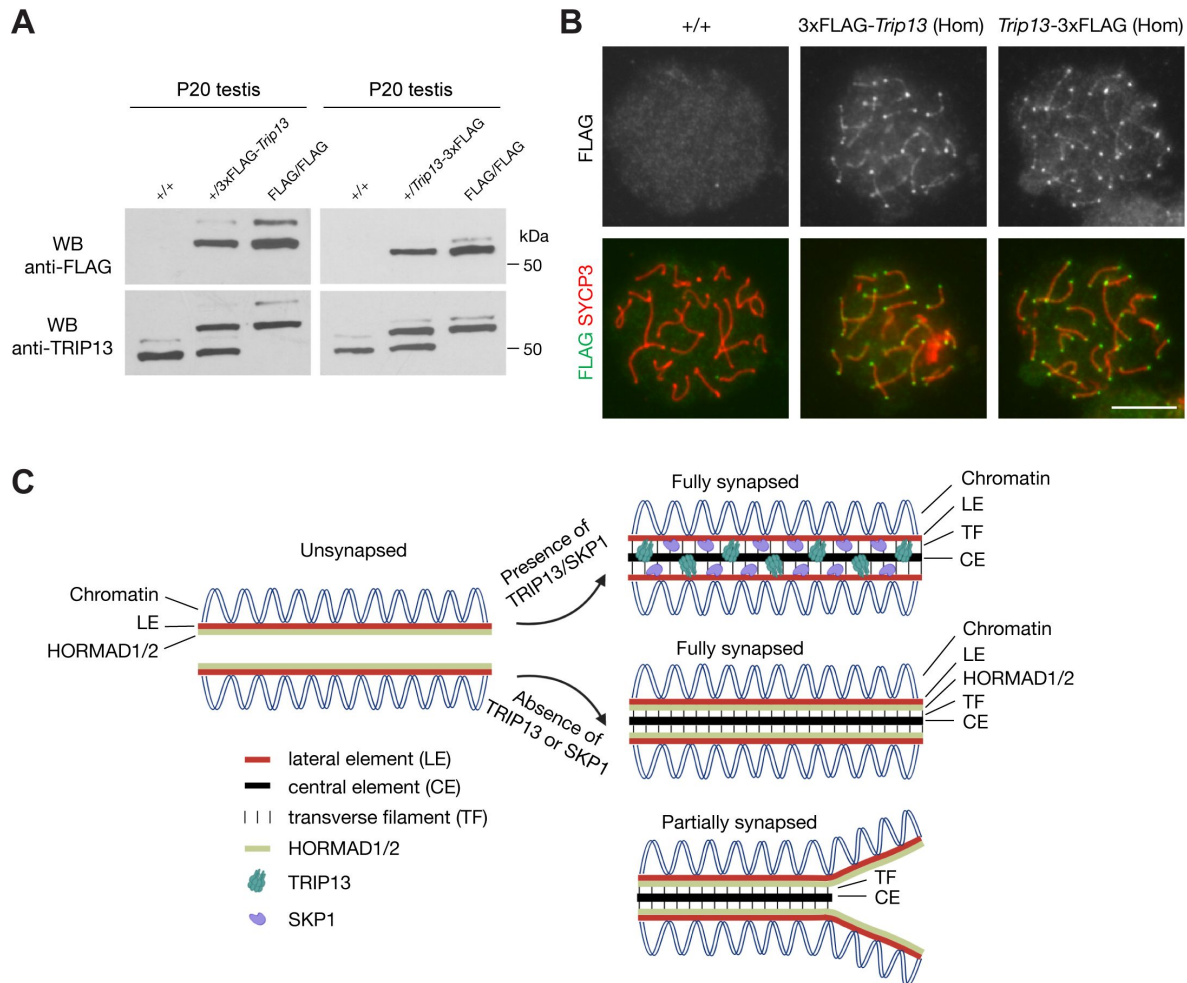
To identify TRIP13-associated proteins in testis, we performed immunoprecipitation (IP) using 3×FLAG-TRIP13, TRIP13-3×FLAG, and wild type (no tag) testicular protein extracts with anti-FLAG monoclonal antibody. The immunoprecipitated proteins were eluted with FLAG peptides and subjected to mass spectrometry for protein identification. As expected, TRIP13 had more peptides in tagged TRIP13 IP than wild type (**Table 1** [↗](#)). HORMAD2, a known TRIP13 substrate protein, was also enriched in tagged TRIP13 IP (**Table 1** [↗](#)). Intriguingly, a large number of RNA-binding proteins involved in RNA splicing were highly enriched in tagged TRIP13 IP: DDX46, PUF60, RBM25, RBM39, U2AF1, U2AF2, and SRSF11 (**Table 1** [↗](#)). The biological relevance of these RNA-binding proteins to TRIP13 function remains unknown. We cannot exclude the possibility that identification of these RNA splicing factors could be immunoprecipitation artifacts.

## Discussion

TRIP13 and its orthologue Pch2 play an evolutionarily conserved role in depletion of meiosis-specific HORMA domain proteins from the synaptonemal complex in many species including yeast, plant, and mouse. In meiocytes, HORMA domain proteins are associated with unsynapsed chromosome axes and become depleted from the SC upon synapsis (**Figure 7C** [↗](#)). In mice with *Trip13* hypomorphic gene trap alleles, HORMAD1 and HORMAD2 persist on the synapsed SCs at the pachytene stage (Wojtasz et al., 2009 [↗](#)). In this study, we confirmed the abnormal persistence of HORMAD1 and HORMAD2 in *Trip13*-null spermatocytes (**Figure 7C** [↗](#)). In budding yeast, deficiency of Pch2 leads to accumulation of Hop1 on the SC (Joshi et al., 2009 [↗](#); Subramanian et al., 2016 [↗](#)). In *Arabidopsis*, Pch2 remodels the HORMA domain protein ASY1 (Balboni et al., 2020 [↗](#); Yang et al., 2020 [↗](#)). Pch2 has been shown to localize to synapsed SC in budding yeast (Joshi et al., 2009 [↗](#)), *C. elegans* (Deshong et al., 2014 [↗](#); Russo et al., 2023 [↗](#)), *Arabidopsis* (Lambing et al., 2015 [↗](#)), and rice (Miao et al., 2013 [↗](#)), but not in mice. In mouse meiocytes, TRIP13 was known to localize to telomeres (Gomez et al., 2019 [↗](#)). In addition to telomeres, we find that TRIP13 localizes to the SC in early pachytene spermatocytes in mice. In particular, TRIP13 localizes as a single filament between the SC lateral elements. Therefore, TRIP13/Pch2 and HORMA domain proteins have mutually exclusive localization patterns on the meiotic chromosome axes, providing an explanation for the depletion of HORMA domain proteins from the synapsed chromosomes in diverse organisms.

The transverse filaments (TF) are essential for recruitment of Pch2 to the SC. In a budding yeast Zip1 (encoding the TF component) allele with four amino acid substitutions, Pch2 is absent from the SC (Subramanian et al., 2016 [↗](#)). In *C. elegans*, the TF component SYP-1 is essential for localization of Pch-2 on the paired chromosomes (Deshong et al., 2014 [↗](#)). In rice, CRC1 (Pch2 orthologue) and CEP1 (TF protein) localize to the SC in an inter-dependent manner (Miao et al., 2013 [↗](#)). In *Arabidopsis*, ZYP1 (TF component) recruits Pch2 to the SC (Yang et al., 2022 [↗](#)). In the yeast two-hybrid assay, rice CRC1 interacts with CEP1, *Arabidopsis* ZYP1 interacts with Pch2, and mouse SYCP1 (TF protein) interacts with TRIP13 (Miao et al., 2013 [↗](#); Yang et al., 2022 [↗](#)). Unlike in other species, loss of SYCP1 (TF protein) in mouse leads to a complete failure in chromosomal synapsis and thus its role in recruitment of TRIP13 to the synapsed SC can not be directly addressed (de Vries et al., 2005 [↗](#)). Our findings on the localization of mouse TRIP13 to the SC on the sister chromatids in *Rec8*-deficient spermatocytes and the short SC stretches in *Sycp2* mutant spermatocytes demonstrate that TRIP13 recruitment is independent of SC axial element proteins REC8 and SYCP2 but support the possibility that SYCP1 could recruit TRIP13 to the synapsed SC. These findings lead to the following model: the TF protein recruits Pch2/TRIP13 to the SC upon synapsis, which in turn evicts HORMA domain proteins from the synapsed regions (**Figure 7C** [↗](#)).





**Figure 7.**

FLAG-tagged TRIP13 proteins localize correctly and are functional. (A) Western blot analysis of tagged and untagged TRIP13 proteins in testes from P20 wild type (no tag), heterozygous tagged, and homozygous tagged males. (B) Immunofluorescence of FLAG-tagged TRIP13 in pachytene spermatocytes from P20 homozygous testes. N-terminal tag, 3xFLAG-*Trip13*; C-terminal tag, *Trip13*-3xFLAG. Scale bar, 10  $\mu$ m. (C) A schematic illustration of TRIP13, SKP1, HORMAD1/2, and the synaptonemal complex. Relative locations of TRIP13 and SKP1 within the SC are depicted. HORMAD1/2 are retained in synapsed regions in *Trip13*-deficient or *Skp1*-deficient spermatocytes.

| Protein | Number of peptides in anti-FLAG IP |                         |                  | M.W. (kD) |
|---------|------------------------------------|-------------------------|------------------|-----------|
|         | 3×FLAG-TRIP13<br>testis            | TRIP13-3×FLAG<br>testis | Wild type testis |           |
| TRIP13  | 28                                 | 36                      | 4                | 51        |
| HORMAD2 | 21                                 | 10                      | 7                | 39        |
| DDX46   | 32                                 | 6                       | 0                | 117       |
| PUF60   | 24                                 | 14                      | 0                | 59        |
| RBM25   | 19                                 | 3                       | 2                | 100       |
| RBM39   | 20                                 | 10                      | 1                | 59        |
| U2AF1   | 11                                 | 4                       | 1                | 28        |
| U2AF2   | 15                                 | 9                       | 3                | 54        |
| SRSF11  | 9                                  | 4                       | 0                | 53        |
| SPATA5  | 26                                 | 27                      | 7                | 97        |
| AP3D1   | 17                                 | 6                       | 2                | 135       |
| PRPF40A | 16                                 | 3                       | 0                | 108       |
| UGP2    | 15                                 | 3                       | 0                | 57        |

**Table 1.**

**List of proteins from testis identified by co-immunoprecipitation and mass spectrometry.**

Both TRIP13 and SKP1 are required for removal of HORMA domain proteins from the synapsed SC (Guan et al., 2020 [DOI](#); Wojtasz et al., 2009 [DOI](#)). However, the relationship of TRIP13 and SKP1 is unknown. SKP1 localizes to the synapsed SC more extensively than TRIP13. While TRIP13 is only detected on the synapsed SC in early pachytene spermatocytes (**Figure 1** [DOI](#)), SKP1 localizes to the synapsed SC in zygotene, all stages of pachytene, and diplotene spermatocytes (Guan et al., 2020 [DOI](#)). The localization of TRIP13 and SKP1 within the SC is different: TRIP13 is on CE/TF regions but SKP1 on LEs. While the global level of TRIP13 is reduced in *Skp1*-deficient testes, the localization of TRIP13 and SKP1 to the synapsed SC is independent (**Figure 6D** [DOI](#)) (Guan et al., 2020 [DOI](#)). The SCF ubiquitin E3 ligase complex targets HORMAD1 for ubiquitination and degradation in transfected HEK293T cells (Guan et al., 2022 [DOI](#)). Loss of TRIP13 leads to accumulation of HORMAD proteins only to the synapsed SC in pachytene cells. In contrast, depletion of SKP1 causes accumulation of HORMA domain proteins, particularly HORMAD1, on both unsynapsed and synapsed chromosome axes from leptotene through diplotene meiocytes. Therefore, TRIP13 and SKP1 might regulate different pools of HORMAD proteins in meiocytes (**Figure 7C** [DOI](#)).

In many organisms including mouse and human, centromeres are the last regions to synapse (Bisig et al., 2012 [DOI](#); Brown et al., 2005 [DOI](#); Qiao et al., 2012 [DOI](#)). The reason for this is unknown, but it might be related to the necessity to suppress deleterious non-homologous recombination at centromere and pericentromeric regions, which consist of minor and major satellite repeats respectively. The centromere was proposed to exert an inhibitory effect on synapsis in human cells (Brown et al., 2005 [DOI](#)). The inhibition needs to be relieved to achieve full synapsis, since unsynapsis at centromeres is expected to trigger the meiotic checkpoint, leading to meiotic arrest. To date, only two mouse mutants (*Skp1* and *Trip13*) exhibit unsynapsis preferentially at the centromeric end in pachytene-like cells. In addition, only these two mouse mutants display abnormal accumulation of HORMAD proteins on the synapsed SC. Intriguingly, TRIP13 localizes to telomeres in both spermatocytes and oocytes. The centromere is close to one of the telomeres in mouse. Thus, the preferential centromeric end asynapsis in *Trip13* or *Skp1*-deficient meiocytes could be related to abnormal persistence of HORMAD proteins. However, the underlying molecular mechanism warrants further investigation.

We find that TRIP13 is a dosage-dependent regulator of meiosis. The *Trip13*<sup>+/-</sup> mice displayed reduced testis weight, reduced sperm count, and meiotic defects but these defects were less severe than the *Trip13*<sup>-/-</sup> mice. The disease phenotypes in humans also appear to be influenced by the *TRIP13* dosage (Yost et al., 2017 [DOI](#); Zhang et al., 2020 [DOI](#)). The dosage-dependent phenotypic variation has been reported in other mouse meiotic mutants such as *Tex11*, *Meiob*, and *Rnf212*. TEX11 localizes to meiotic chromosomes as foci and regulates crossover formation and chromosome synapsis (Yang et al., 2008 [DOI](#)). The TEX11 protein levels correlate with genome-wide recombination rates in mice (Yang et al., 2015 [DOI](#)). MEIOB, a meiosis-specific ssDNA-binding protein, is essential for meiotic recombination (Luo et al., 2013 [DOI](#); Souquet et al., 2013 [DOI](#)). In mice with different combinations of *Meiob* alleles, MEIOB protein levels directly correlate with the severity of meiotic defects (Guo et al., 2020 [DOI](#)). Sequence variants in human RNF212 are associated with variations in genome-wide recombination rates (Chowdhury et al., 2009 [DOI](#); Kong et al., 2008 [DOI](#)). In mouse, RNF212 forms foci on meiotic chromosomes and regulates crossover formation in a dosage-dependent manner (Reynolds et al., 2013 [DOI](#)). Therefore, dosage sensitivity appears to be a common feature of many regulators of meiosis.

## Materials and Methods

### Ethics statement

Mice were maintained and used for experimentation according to the protocol approved by the Institutional Care and Use Committee of the University of Pennsylvania.

## Mouse strains

The cryopreserved sperm from *Trip13*<sup>+/-</sup> males were obtained from the Mutant Mouse Resource and Research Center (MMRCC) at UC Davis (*Trip13*<sup>tm1.1(KOMP)Vlcg/JMmucd</sup>, MMRRC\_050223-UCD). Genotyping of wild type and knockout *Trip13* alleles was performed by separate PCR reactions using tail genomic DNA. Other mutant mouse lines used in this study were previously generated: *Hormad1*, *Sycp2*, and *Rec8* (Guan et al., 2020 [DOI](#); Shin et al., 2010 [DOI](#); Yang et al., 2006 [DOI](#)). Genotyping PCR primer sequences are listed in [Table 2](#) [DOI](#).

## Generation of 3×FLAG-*Trip13* and *Trip13*-3×FLAG knockin mouse strains

The 3×FLAG-knockin mouse strains were generated by the CRISPR-Cas9 mediated genome editing approach. To generate N-terminally tagged knockin mice, one single-guide RNA (sgRNA) was designed to target the first exon of the mouse *Trip13* ([Figure 7 – figure supplement 1A](#) [DOI](#)). A ssDNA repair template was created using the sequence surrounding the start codon as 5' and 3' homology arms. The knockin template itself included the sequence for the 3×FLAG epitope and a three-residue linker sequence. The C-terminally tagged strain was generated similarly ([Figure 7 – figure supplement 1B](#) [DOI](#)). The guide RNA targeted exon 13 containing the stop codon. The ssDNA template contained the FLAG-encoding sequences and homology sequences flanking the insertion site. sgRNA sequences and ssDNA template sequences were shown in [Table 2](#) [DOI](#).

For the sgRNA, the oligo was phosphorylated, annealed, and cloned to PX330 plasmid (Addgene, Waterton, MA). After *in vitro* transcription with the MEGAshortscript T7 Kit (AM1354, Invitrogen) and purification with the MEGAclean Transcription Clean-Up Kit (AM1908, Invitrogen), a mixture of Cas9 mRNA [100ng/μl; Trilink, catalog number L-7206] + 0.5 μl of the sgRNA (35 ng/μl) + 100ng/μl of ssDNA template] was prepared and injected into zygotes. The injected zygotes were cultured in KSOM medium at 37°C in a 5% CO<sub>2</sub> incubator until the two-cell stage. The two-cell embryos were transferred into oviducts of 0.5-day post-coitum pseudopregnant ICR foster mothers. Founder mice were bred to wild type mice to obtain germline transmission. The N-terminal 3×FLAG allele and the C-terminal 3×FLAG allele were PCR amplified and sequenced to confirm the insertion. PCR genotyping primers are listed in [Table 2](#) [DOI](#).

## Production of anti-MAJIN antibodies

The short isoform of mouse MAJIN (amino acids 1-124; XM\_036161650.1 and XP\_036017543.1) was expressed as a 6xHis-MAJIN recombinant protein in *E. coli* using the pQE-30 vector. The recombinant protein was affinity purified with the Ni-NTA agarose. Two guinea pigs were immunized at Cocalico Biologicals Inc. (Reamstown, PA), resulting in two antisera (UP-GP140 and UP-GP141). Both anti-sera were used for immunofluorescence of nuclear spreads of spermatocytes.

## Histological, immunofluorescence, and surface nuclear spread analyses

For histology, testes or ovaries were fixed in Bouin's solution at room temperature overnight, embedded with paraffin, and sectioned at 8 μm. Sections were stained with hematoxylin and eosin. For immunofluorescence analysis, testes were fixed in 4% paraformaldehyde (in 1×PBS) overnight at 4°C, dehydrated in 30% sucrose (in 1×PBS) overnight, and sectioned at 8 μm in a cryostat. Surface nuclear spread analysis was described before (Peters et al., 1997 [DOI](#)). Briefly, testicular tubules or ovarian tissues were soaked in hypotonic treatment buffer (30 mM Tris, 50 mM sucrose, 17 mM trisodium citrate dihydrate, 5 mM EDTA, 0.5 mM DTT, 1 mM PMSF). Then the cells were suspended in 100 mM sucrose and spread by physical disruption on "PTFE" printed slides that were previously soaked with paraformaldehyde solution containing Triton X-100 and sodium borate. Antibodies used for immunostaining are listed in [Table 3](#) [DOI](#).

| Genotyping primers    |                          |                                                                                                                                                                                                                              |                  |
|-----------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Allele                | Forward                  | Reverse                                                                                                                                                                                                                      | product (bp)     |
| <i>Trip13</i> WT      | GCCCTTAGCCAAGGTGGAT      | TCCTTGACCCCCTAATTGAC                                                                                                                                                                                                         | 645              |
| <i>Trip13</i> KO      | ACTTGCTTTAAAAAACCTCCCACA | CAGAAAGCAACTGCTCCCTTCTAGC                                                                                                                                                                                                    | 731              |
| <i>Sycp2</i> WT       | AGATGAGGGCATATCACCGA     | TAAGCACACTCACCATCTCC                                                                                                                                                                                                         | 400              |
| <i>Sycp2</i> KO       | GCATGTTATCAACCTTATCCCT   | CCTACCGGTGGATGTGGAATGTGTG                                                                                                                                                                                                    | 300              |
| <i>Rec8</i> WT        | AGCAGAGTCGAAGAAGGCCTCTTG | CAGATGGTGGCGAAGCAGCCTGT                                                                                                                                                                                                      | 426              |
| <i>Rec8</i> KO        | AGCAGAGTCGAAGAAGGCCTCTTG | TTGCTCAGGGGAATTTGGGTC                                                                                                                                                                                                        | 212              |
| <i>Hormad1</i> WT     | TCAAGACCAACCTGGGCTAC     | CCATGTGGGTTGTAGGGAGT                                                                                                                                                                                                         | 196              |
| <i>Hormad1</i> KO     | TCAAGACCAACCTGGGCTAC     | GGGGAACCTCCTGACTAGGG                                                                                                                                                                                                         | ~400             |
| 3×FLAG- <i>Trip13</i> | CCTACATCGGAGAAGGCTGT     | TTCATGTCAGGCTGTTTCAGG                                                                                                                                                                                                        | WT:348<br>KI:426 |
| <i>Trip13</i> -3×FLAG | GCCCCACTAAAGCACAAGTC     | ACAGGCTTGAGTCAGGATGG                                                                                                                                                                                                         | WT:405<br>KI:471 |
| Genome editing        |                          |                                                                                                                                                                                                                              |                  |
|                       | guide RNA                | HDR ssDNA                                                                                                                                                                                                                    |                  |
| 3×FLAG- <i>Trip13</i> | ATTCCCTCGGCTCCCGGCGG     | TCGGAGAAGGCTGTCGCACAGGGCGCAGGGAGGCGA<br>CCGCGGCCTCACTCCGGCGGCATTCCCTCGGCTCCCG<br>GCGGCAGCGCCATGGGTGACTACAAAGACCATGACG<br>GTGATTATAAAGATCATGACATCGATTACAAGGATGA<br>CGATGACAAGGGAAGCGGAGACGAGGCGGTGGGCGA<br>CCTGAAGCAAGCGCTTCC |                  |
| <i>Trip13</i> -3×FLAG | AAGCCATAGATATGGATGTC     | AGGGTTTCCTCCAGGCCCTATCTCTGGCAGTGGACAA<br>ACAGTTTGAGGAGAAAAAGAACTTTCAGCTTATGTT<br>GACTACAAAGACCATGACGGTGATTATAAAGATCATG<br>ACATCGATTACAAGGATGACGATGACAAGTGATCCAA<br>GACATCCATATCTATGGCTTTCATGGACAAGTAGGA<br>GGTGATACCGTCTAC   |                  |

**Table 2.**

**Sequences of genotyping PCR primers, sgRNA, and ssDNA templates**



| Reagent                       | Dilution            | Source                                             | Catalog # or reference             |
|-------------------------------|---------------------|----------------------------------------------------|------------------------------------|
| Mouse anti-ACTB mab           | 1:2000              | Sigma                                              | Cat# A5441; RRID: AB_476744        |
| Human anti-centromere (CREST) | 1:500               | Antibodies Incorporated                            | Cat# 15-234; RRID: AB_2687472      |
| Rabbit anti-SYCP1             | 1:300               | Abcam                                              | Cat# ab15090; RRID: AB_301636      |
| Guinea pig anti-SYCP2         | 1:150               | Custom made                                        | (Yang et al., 2006)                |
| Mouse anti-SYCP3 mab          | 1:500               | Abcam                                              | Cat# ab97672; RRID: AB_10678841    |
| Rabbit anti-SYCP3             | 1:500 IF; 1:2000 WB | ProteinTech Group                                  | Cat# 23024-1-AP; RRID: AB_11232426 |
| Rabbit anti-HORMAD1           | 1:300               | ProteinTech Group                                  | Cat# 13917-1-AP; RRID: AB_2120844  |
| Rabbit anti-HORMAD2           | 1:500               | Custom made, a gift from Toth lab                  | (Wojtasz et al., 2009)             |
| Rabbit anti-REC114            | 1:100               | Custom made, a gift from Keeney Lab                | (Boekhout et al., 2019)            |
| Rabbit anti-REC8              | 1:200               | Custom made, a gift from Mengcheng Luo lab         | N/A                                |
| Mouse anti-FLAG mab           | 1:300IF; 1:4000WB   | Sigma                                              | Cat# F3165, RRID: AB_259529        |
| Rabbit anti-SKP1              | 1:200 IF; 1:1000 WB | Cell Signaling                                     | Cat# 12248S, RRID: AB_2754993      |
| Rabbit anti-TRIP13            | 1:150 IF; 1:1000 WB | ProteinTech Group                                  | 19602-1-AP                         |
| Rabbit anti-SYCE1             | 1:200 IF; 1:1000 WB | Custom made                                        | N/A                                |
| Mouse anti-SYCP1 mab          | 1:200 IF; 1:1000 WB | Custom made, 1:200; a gift from Chritster Hoog lab | N/A                                |
| Rabbit anti-CENP-C            | 1:500               | Custom made, a gift from Y. Watanabe               | (Kim et al., 2015)                 |
| Mouse anti-TRF1 mab           | 1:100               | Sigma                                              | T1948                              |
| Guinea pig anti- MAJIN        | 1:100               | Custom made; this study                            | N/A                                |

**Table 3.**

### List of primary antibodies

## Imaging

Histological images were captured on the Leica DM5500B microscope with a DFC450 digital camera (Leica Microsystems, Wetzlar, Germany). Most immunolabeled chromosome spread images were taken on the Leica DM5500B microscope with an ORCA Flash4.0 digital monochrome camera (Hamamatsu Photonics, Bridgewater, NJ). Confocal microscopy of immunolabeled chromosome spreads was performed on a Leica SP5 II confocal (Leica Microsystems, Wetzlar, Germany) with an 100× (1.46 NA) oil immersion objective lens. Images were deconvolved with Huygens Essential deconvolution software (Scientific Volume Imaging B.V., Hilversum, Netherlands).

## Western blot analysis

Testes were homogenized in the lysis buffer (50 mM Tris-HCl, pH 8.0, 150 mM NaCl, 1% Trion X-100, 0.5% sodium deoxycholate, 5 mM MgCl<sub>2</sub>, and 1 mM DTT supplemented with 1 mM PMSF). 40 µg of protein samples were resolved by SDS-PAGE, transferred onto PDVF membranes, and immunoblotted with primary antibodies ([Table 3](#) [↗](#)).

## Acknowledgements

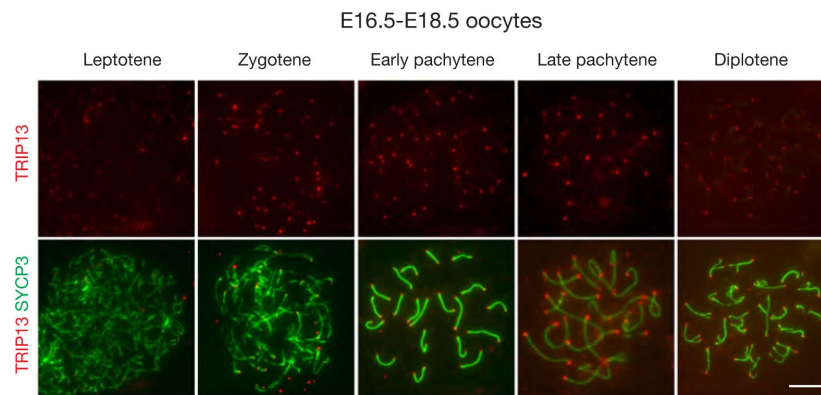
We thank Gordon Ruthel at the PennVet Imaging Core for help with super-resolution microscopy, Hsin Yao Tang and Thomas Beer at Wistar Proteomics Core for help with mass spectrometry, Christer Hoog for SYCP1 antibody, Attila Toth for HORMAD2 antibody, Scott Keeney for anti-REC114 antibody, Yoshinori Watanabe and Takashi Akera for CENP-C antibody.

## Competing interests

The authors declare that no competing interests exist.

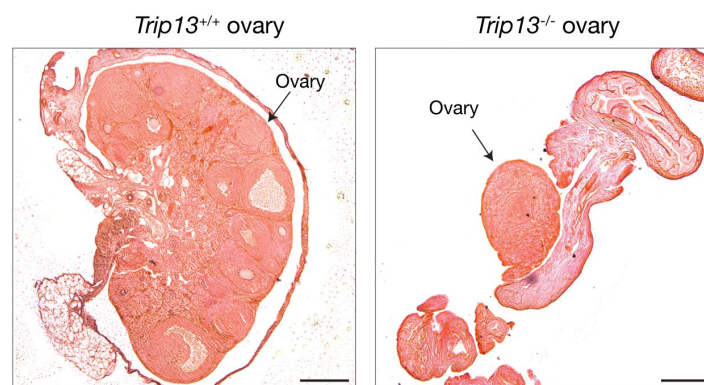
## Funding

This work was supported by National Institutes of Health/National Institute of Child Health and Human Development grants T32HD083185 (JYC), R01HD109253 (PJW), and P50HD068157 (PJW).



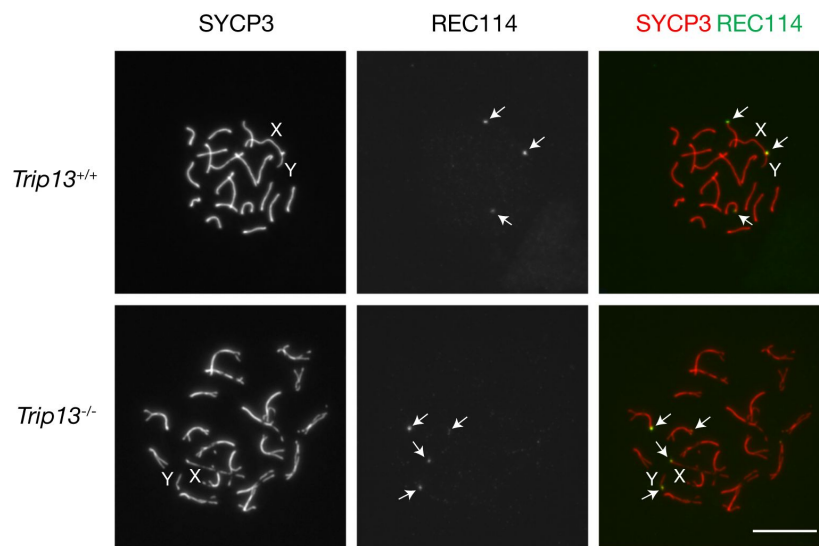
**Figure 1 - figure supplement 1.**

Immunofluorescent analysis of TRIP13 in spread nuclei of oocytes from embryonic day 16.5 (E16.5) and E18.5 female embryos. Scale bars, 10  $\mu$ m.



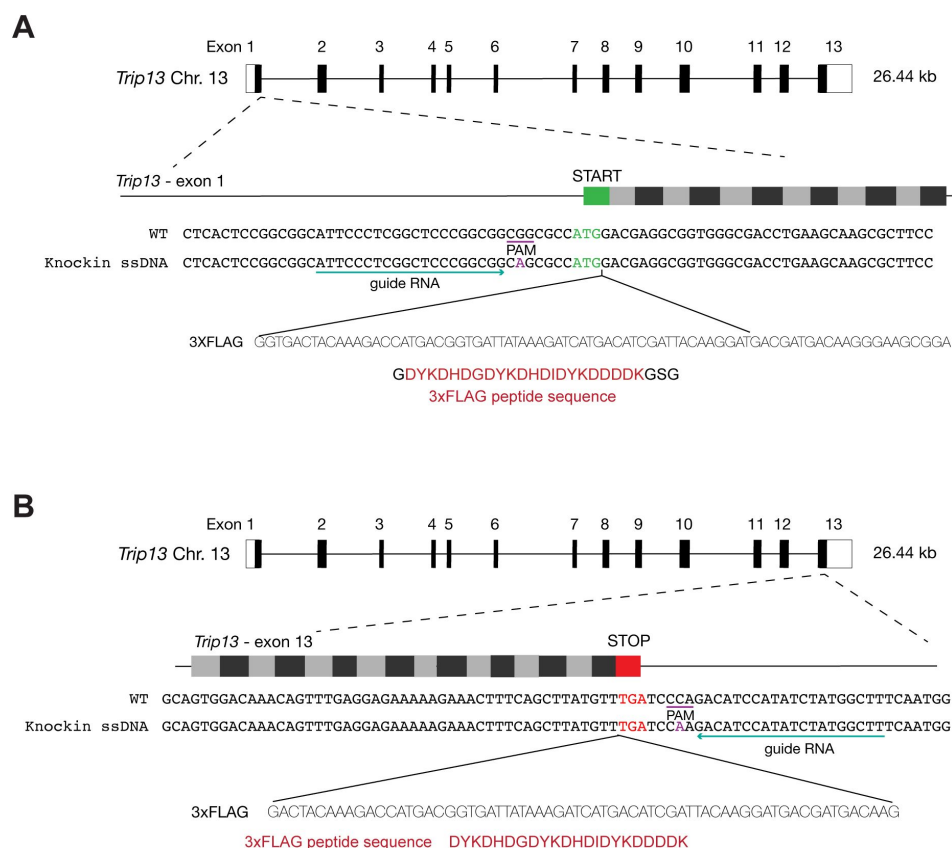
**Figure 2 - figure supplement 1.**

Histological analysis of ovaries from adult (8-week) wild type and *Trip13*<sup>-/-</sup> females. Scale bars, 200  $\mu$ m.



**Figure 4 - figure supplement 1.**

Immunofluorescent analysis of REC114 in spread nuclei of spermatocytes from P28 wild type (*Trip13*<sup>+/+</sup>) and *Trip13*<sup>-/-</sup> testes. Wild type, early pachytene; *Trip13*<sup>-/-</sup>, pachytene-like. REC114 foci are indicated by arrows. Scale bar, 10  $\mu$ m.



**Figure 7 - figure supplement 1.**

Generation of two *Trip13* FLAG tagged mouse lines. (A) Illustration of the *Trip13* gene structure with the 3xFLAG tag at the N terminus. The guide RNA sequence is underlined in green. The single strand DNA (ssDNA) oligo template (200 nt) contains the 3xFLAG-encoding sequence. The PAM site is underlined in purple. One base in the PAM site is mutated in the ssDNA oligo template to prevent cutting of the template strand (in purple). The position of 3xFLAG in-frame insertion is designated (^) and occurs just after the endogenous start codon. Filled bars, coding regions; Open bars, 5' or 3' UTRs. (B) Illustration of the *Trip13* gene structure with the 3xFLAG tag at the C terminus. The PAM cut site and guide RNA were on the reverse strand, but for clarity, the forward strand sequence is shown. The 3xFLAG was inserted just before the endogenous stop codon.



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## Article and author information

### Jessica Y. Chotiner

Department of Biomedical Sciences, University of Pennsylvania School of Veterinary Medicine, Philadelphia, PA 19104, USA

### N. Adrian Leu

Department of Biomedical Sciences, University of Pennsylvania School of Veterinary Medicine, Philadelphia, PA 19104, USA

### Fang Yang

Department of Biomedical Sciences, University of Pennsylvania School of Veterinary Medicine, Philadelphia, PA 19104, USA

### Isabella G. Cossu

Department of Biomedical Sciences, University of Pennsylvania School of Veterinary Medicine, Philadelphia, PA 19104, USA

**Yongjuan Guan**

Department of Biomedical Sciences, University of Pennsylvania School of Veterinary Medicine, Philadelphia, PA 19104, USA, College of Life Sciences, Capital Normal University, Beijing, China

**Huijuan Lin**

Department of Biomedical Sciences, University of Pennsylvania School of Veterinary Medicine, Philadelphia, PA 19104, USA

**P. Jeremy Wang**

Department of Biomedical Sciences, University of Pennsylvania School of Veterinary Medicine, Philadelphia, PA 19104, USA

**For correspondence:** [pwang@vet.upenn.edu](mailto:pwang@vet.upenn.edu)

ORCID iD: [0000-0003-2311-4089](https://orcid.org/0000-0003-2311-4089)

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**Editors**

Reviewing Editor

**Wei Yan**

The Lundquist Institute, Torrance, United States of America

Senior Editor

**Wei Yan**

The Lundquist Institute, Torrance, United States of America

**Reviewer #1 (Public Review):****Summary:**

TRIP13/Pch2 is a conserved essential regulator of meiotic recombination from yeast to humans. In this manuscript, the authors generated TRIP13 null mice and Flag-tagged TRIP13 knock-in mice to study its role in meiosis. They demonstrate that TRIP13 regulates MORF domain proteins and is essential for meiotic completion and fertility. The main impact of this manuscript is its clarification of the *in vivo* function of TRIP13 during mouse meiosis and previously unrecognized role as a dose-sensitive regulator of meiosis.

**Strengths:**

Two previously reported Trip13 mutations in mice are both hypomorphic alleles with distinct phenotypes, precluding a conclusion on its function. This study for the first time generated the TRIP13 null mice, definitively revealed the function of TRIP13 in meiosis. The authors also show novel localization of TRIP13 at SC and its independence from the axial element components. The finding of dose-sensitive regulation of meiosis by TRIP13 has implication in understanding human meiosis and disease phenotypes.

The results support the main conclusions and advance the understand of meiosis in the germline.

<https://doi.org/10.7554/eLife.92195.2.sa1>



## Reviewer #2 (Public Review):

### Summary and Strengths:

In this manuscript, Chotiner and colleagues demonstrated the localization of TRIP13 and clarified the phenotypes of Trip13-null mice in mouse meiosis. The meiotic phenotypes of Trip13 have been well characterized using the hypomorph alleles in the literature. However, the null phenotypes have not been examined, and the localization of TRIP13 was not clearly demonstrated. The study fills these important knowledge gaps in the field. The demonstration of TRIP13 localization to SC in mice provides an explanation of how HOMRA domain proteins are evicted from SC in diverse organisms. This conclusion was confirmed in both IF and TRIP13-tagged Tg mice. Further, the phenotypes of Trip13-null mice are very clear. The manuscript is well crafted, and the discussion section is well organized and comprehends the topic in the field. All in all, the manuscript will provide important knowledge in the field of meiosis.

### Weaknesses:

The heterozygous phenotypes demonstrate that TRIP13 is a dosage-sensitive regulator of meiosis. In relation to this conclusion, as summarized in the discussion section, other mutants defective in meiotic recombination showed dosage-sensitive phenotypes. However, the authors did not examine meiotic recombination in the Trip13-null mice.

<https://doi.org/10.7554/eLife.92195.2.sa0>

## Author Response

The following is the authors' response to the original reviews.

### Public Reviews:

#### Reviewer #1 (Public Review):

##### Summary:

*TRIP13/Pch2 is a conserved essential regulator of meiotic recombination from yeast to humans. In this manuscript, the authors generated TRIP13 null mice and Flag-tagged TRIP13 knock-in mice to study its role in meiosis. They demonstrate that TRIP13 regulates MORMA domain proteins and is essential for meiotic completion and fertility. The main impact of this manuscript is its clarification of the in vivo function of TRIP13 during mouse meiosis and its previously unrecognized role as a dose-sensitive regulator of meiosis.*

##### Strengths:

*Two previously reported Trip13 mutations in mice are both hypomorphic alleles with distinct phenotypes, precluding a conclusion on its function. This study for the first time generated the TRIP13 null mice, definitively revealing the function of TRIP13 in meiosis. The authors also show the novel localization of TRIP13 at SC and its independence from the axial element components. The finding of dose-sensitive regulation of meiosis by TRIP13 has implications in understanding human meiosis and disease phenotypes.*

##### Weaknesses:

*This manuscript would be more impactful if more mechanistic advancements could be made. For example, the authors could follow up with one of the new interactors identified by MS to offer new insight into the molecular function of TRIP13.*

We agree that it would be interesting to follow up on new candidate interactors but think that it would be more feasible to follow up on them in future studies.

**Reviewer #2 (Public Review):**

*Summary and Strengths:*

*In this manuscript, Chotiner and colleagues demonstrated the localization of TRIP13 and clarified the phenotypes of Trip13-null mice in mouse meiosis. The meiotic phenotypes of Trip13 have been well characterized using the hypomorph alleles in the literature. However, the null phenotypes have not been examined, and the localization of TRIP13 was not clearly demonstrated. The study fills these important knowledge gaps in the field. The demonstration of TRIP13 localization to SC in mice provides an explanation of how HOMRA domain proteins are evicted from SC in diverse organisms. This conclusion was confirmed in both IF and TRIP13-tagged Tg mice. Further, the phenotypes of Trip13-null mice are very clear. The manuscript is well crafted, and the discussion section is well organized and comprehends the topic in the field. All in all, the manuscript will provide important knowledge in the field of meiosis.*

*Weaknesses:*

*The heterozygous phenotypes demonstrate that TRIP13 is a dosage-sensitive regulator of meiosis. In relation to this conclusion, as summarized in the discussion section, other mutants defective in meiotic recombination showed dosage-sensitive phenotypes. However, the authors did not examine meiotic recombination in the Trip13-null mice.*

Meiotic recombination was extensively characterized in Trip13 severe hypomorph mutants in two previous studies: gamma-H2AX, BLM, BRCA1, ATR, RPA, RAD51, DMC1, MLH1 (Li and Schimenti, 2007; Roig et al., 2010). All the meiotic defects in our Trip13-null mice were also present in Trip13 severe hypermorph mutants: meiotic arrest, defects in chromosomal synapsis, asynapsis at chromosomal ends, and accumulation of HORMAD1/2 on the SC axis. Therefore, the defects in meiotic recombination in Trip13-null mice are expected to be similar to those in Trip13 severe hypermorph mutants and thus we did not examine the proteins involved in meiotic recombination in the Trip13-null mutant.

**Reviewer #3 (Public Review):**

*Summary:*

*The authors perform a thorough examination of the phenotypes of a newly generated Trip13 null allele in mice, noting defects in chromosome synapsis and impact on localization of other key proteins (namely HORMADs) on meiotic chromosomes. The vast majority of data confirms observations of several prior studies of Trip13 alleles (moderate and severe hypomorphs). The original or primary aims of the study aren't clear, but it can be assumed that the authors wanted to better study the role of this protein in evicting HORMADs upon synapsis by studying phenotypes of mutants and better characterizing TRIP13 localization data (which they find localizes to the central element of synapsed chromosomes using a new epitope-tagged allele). Their data confirm prior reports and are consistent with localization data of the orthologous Pch2 protein in many other organisms.*

*Strengths:*

*The quality of data is high. Probably the most important data the authors find is that TRIP13 is localized along the CE of synapsed chromosomes. However, this was not*

*unexpected because PCH2 is also similarly localized. Also, the authors use a clear null (deletion allele), whereas prior studies used hypomorphs.*

**Weaknesses:**

*There is limited new data; most are confirmatory or expected (i.e., SC localization), and thus the impact of this report is not high. The claim that TRIP13 "functions as a dosage-sensitive regulator of meiosis" is exaggerated in my opinion. Indeed, the authors make the observation that hets have a phenotype, but numerous genes have haploinsufficient phenotypes. In my opinion, it is a leap to extrapolate this to infer that TRIP13 is a "regulator" of meiosis. What is the definition of a meiosis regulator? Is it at the apex of the meiosis process, or is it a crucial cog of any aspect of meiosis?*

TRIP13 is not haploinsufficient, as Trip13 heterozygotes were still viable and fertile (albeit with defects in meiosis). TRIP13 is an ATPase and changes the conformation of meiosis-specific proteins such as HORMAD proteins. TRIP13 is essential for meiosis and its mutations cause defects in both meiotic recombination and chromosomal synapsis. Reviewer 1 stated that "TRIP13/Pch2 is a conserved essential regulator of meiotic recombination from yeast to humans". Therefore, we feel that TRIP13 can be called a regulator of meiosis.

**Reviewer #1 (Recommendations For The Authors):**

*A schematic illustration of SC structure, the components involved, and the main finding, would be helpful for readers to better understand the advancement made by this study.*

We have now added a schematic illustration in a new panel - Figure 7C.

*Fig. 1B, the stage with diplotene cells should be XII.*

The pachytene cells (Pac) were mis-labelled as diplotene cells. Corrected.

*Fig. 1C, color mislabeled.*

Corrected.

**Reviewer #2 (Recommendations For The Authors):**

*The manuscript will provide important knowledge in the field of meiosis. I support the publication of this study. I have some suggestions to improve and polish the manuscript.*

**Major points:**

*(1) The heterozygous phenotypes demonstrate that TRIP13 is a dosage-sensitive regulator of meiosis. In relation to this conclusion, as summarized in the discussion section, other mutants defective in meiotic recombination showed dosage-sensitive phenotypes. Given the function of HORMAD1 in meiotic recombination, it would be informative if the authors could examine how major makers of meiotic recombination behave in Trip13-null meiosis.*

Please see our response to Weaknesses from Reviewer #2.

*(2) Relating to the above point, the complete lack of synapsis on the sex chromosomes in the Trip13-null meiosis is impressive. This result raises a question as to whether the pathway to designate XY-obligatory crossover (which can be detected with large foci of ANKRD31 and MEI4/REC114 at PAR) is affected or not. It would be interesting to examine whether the ANKRD31 and MEI4/REC114 foci are present on PAR in Trip13-null meiosis.*

We have performed immunofluorescent analysis of REC114 in spermatocytes. In Trip13-null pachytene-like spermatocytes, X and Y chromosomes are not synapsed. REC114 still formed one focus each on the unsynapsed X and Y chromosomes. We have added this new data in the Results as a new supplementary figure (Figure 4-supplement 1).

*(3) Figure 4 can be improved if there are quantified data for each phenotype. These phenotypes look nearly complete, but it would be informative to show the penetrance of these phenotypes.*

Because some chromosomes have unsynapsed ends, resulting in two centromere or telomere foci, the total number of centromere or telomere foci is always higher in Trip13-null pachytene-like spermatocytes than wild type pachytene spermatocytes. Therefore, we did not count the foci of centromeres and telomeres. Consistently, the centromere and telomere markers localized as expected in both wild type and Trip13-null spermatocytes.

*(4) I am not fully convinced by these photos: "synapsed sister chromatids (Figure 6B)" and "Sycp2-/- spermatocytes formed short stretches of synapsis (Figure 6C)". The authors may try confocal microscopy with super-resolution deconvolution as they did for other data.*

These have been previously demonstrated. The “synapsed sister chromatids (Figure 6B)” were previously demonstrated by confocal microscopy with super-resolution deconvolution (Guan et al., 2020). The short stretches of synapsis in Sycp2-/- spermatocytes was previously demonstrated by electron microscopy (Tripartite SC structure) and SYCP1 immunofluorescence (Yang et al., 2006). We have revised the text by citing the previous evidence and the publications.

*Minor points:*

*(1) Line 19-21: "Loss of TRIP13 leads to meiotic arrest and thus sterility in both sexes. Trip13-null meiocytes exhibit abnormal persistence of HORMAD1 and HOMRAD2 on synapsed SC". These findings confirm the previously reported phenotypes of the Trip13 hypomorph alleles. This information can be added to the abstract. Otherwise, it sounds like these are totally new findings, as written.*

This information is now added to the abstract: “These findings confirm the previously reported phenotypes of the Trip13 hypomorph alleles.”

*(2) The introduction section seems too long and contains unnecessary information. Some molecular details that are not touched in the result section can be deleted (e.g., Line 65-73).*

We would like to keep the molecular details on the two conformation states, as it provides biochemical background on TRIP13-HORMAD interactions.

*(3) Introduction, Line 92. A rationale can be added as to why the authors characterized the Trip13-null allele.*

a rationale has been added as follows: “To determine the effect of complete loss of TRIP13, we characterized Trip13-null mice.”

*(4) Line 205: Typo "TRRIP13". Corrected.*

**Reviewer #3 (Recommendations For The Authors):**

*Just a few recommendations:*

*(1) In my opinion, the title is an overreach. "Regulator" invokes other concepts such as transcription factors.*

Please see our explanation in response to weaknesses from Reviewer #3.

*(2) The first sentence of the results deals with TRIP13 expression in only 3 tissues. The authors might look at more comprehensive RNA-seq data from mice and humans.*

We examined TRIP13 protein expression in 8 mouse tissues by WB and found that TRIP13 protein was abundant in testis but present at a very low level in ovary and liver (Figure 1A). We feel that readers can easily look up the relative transcript levels of Trip13 in more tissues from mice and humans from NCBI database under "Gene".

*(3) The null allele is semi-lethal. Is body size affected? Were the mice abnormal in any other ways, given that TRIP13 has been implicated in other diseases and processes, and is expressed in other tissues (TRIP13 stands for Thyroid receptor interacting protein).*

The body weight of 2-3 month-old males was not significantly different between wild type ( $24.3 \pm 2.8$  g,  $n=5$ ) and Trip13 KO mice ( $22.8 \pm 1.7$  g,  $n=5$ ,  $p=0.3$ , Student's t-Test). We have included the body weight information in the revised manuscript. We didn't observe abnormal somatic defects in the viable Trip13-null mice, nor did the authors report any in the Trip13 hypomorph mutants in two previous studies (Li and Schimenti, 2007; Roig et al., 2010).

*(4) Line 276 : It would be nice to elaborate on the "spatial explanation."*

We meant that TRIP13 localizes to SC while HORMAD proteins are removed from SC upon chromosomal synapsis, thus providing a spatial explanation. However, we have now deleted "spatial".